www.tntextbooks.in

z directions of motion) so that each degree of freedom will get $\frac{1}{2}kT$ *of energy. This is called law of equipartition of energy.*

Average kinetic energy of a monatomic molecule (with f=3) = $3\times\frac{1}{2}kT=\frac{3}{2}kT$

Average kinetic energy of diatomic molecule at low temperature (with f = 5) = $5\times\frac{1}{2}kT=\frac{5}{2}kT$

Average kinetic energy of a diatomic molecule at high temperature (with f =7) = $7\times\frac{1}{2}kT=\frac{7}{2}kT$

Average kinetic energy of linear triatomic molecule (with f = 7) = $7\times\frac{1}{2}kT=\frac{7}{2}kT$

Average kinetic energy of non linear triatomic molecule (with f = 6) = $6\times\frac{1}{2}kT=3kT$

9.4.1 Application of law of equipartition energy in specific heat of a gas

Meyer's relation $C_P - C_V = R$ connects the two specific heats for one mole of an ideal gas.

Equipartition law of energy is used to calculate the value of $C_P - C_V$ and the ratio between them $\gamma = \frac{C_P}{C_V}$. Here γ is called adiabatic exponent.

i) Monatomic molecule

Average kinetic energy of a molecule

$$=\left[\frac{3}{2}kT\right]$$

Total energy of a mole of gas

$$=\frac{3}{2}kT\times N_A=\frac{3}{2}RT$$

For one mole, the molar specific heat at constant volume $C_V=\frac{dU}{dT}=\frac{d}{dT}\left[\frac{3}{2}RT\right]$

$$C_V=\left[\frac{3}{2}R\right]$$

$$C_P=C_V+R=\frac{3}{2}R+R=\frac{5}{2}R$$

The ratio of specific heats,

$$\gamma=\frac{C_P}{C_V}=\frac{\frac{5}{2}R}{\frac{3}{2}R}=\frac{5}{3}=1.67$$

ii) Diatomic molecule

Average kinetic energy of a diatomic molecule at low temperature = $\frac{5}{2}kT$

Total energy of one mole of gas

$$=\frac{5}{2}kT\times N_A=\frac{5}{2}RT$$

(Here, the total energy is purely kinetic)

For one mole Specific heat at constant volume

$$C_V=\frac{dU}{dT}=\left[\frac{5}{2}RT\right]=\frac{5}{2}R$$

But $C_P=C_V+R=\frac{5}{2}R+R=\frac{7}{2}R$

$$\therefore\gamma=\frac{C_P}{C_V}=\frac{\frac{7}{2}R}{\frac{5}{2}R}=\frac{7}{5}=1.40$$

Energy of a diatomic molecule at high temperature is equal to $\frac{7}{2}RT$

$$C_V=\frac{dU}{dT}=\left[\frac{7}{2}RT\right]=\frac{7}{2}R$$

$$\therefore C_P=C_V+R=\frac{7}{2}R+R$$

$$C_P=\frac{9}{2}R$$

Note that the C_V and C_P are higher for diatomic molecules than the mono atomic molecules. It implies that to increase the temperature of